Submodule 40.04: Landmark CNN Architectures: LeNet Se ResNet Tak (Hinglish Guide)

1. Deep Learning Ka Itihaas

Computer Vision ka modern revolution 5 sabse mashhoor architectures ki wajah se aaya:

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1998            2012            2014            2014            2015
LeNet-5 ------> AlexNet ------> VGG-16 -------> GoogLeNet -----> ResNet
(Yann LeCun)    (Krizhevsky)    (Simonyan)      (Inception)     (Kaiming He)
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2. In 5 Landmark Models Ki Mukhya Khoj

1. LeNet-5 (1998, Yann LeCun)

- **Kaam**: Duniya ka pehla safal CNN model jo US banks mein bank checks ke handwritten numbers (MNIST) padhne ke liye use hua.
- **Structure**: 2 Conv layers (5×5) + Average Pooling + 2 Dense layers.

2. AlexNet (2012, Alex Krizhevsky & Geoffrey Hinton)

- **Deep Learning Ka Big Bang**: ImageNet competition ko 10% ke historic margin se jeeta.
- **Badi Inventions**:
    1. Pehli baar model ko **NVIDIA GPUs** par train kiya.
    2. Sigmoid ki jagah **ReLU** use kiya (jisse training $6\times$ tezz ho gayi).
    3. Overfitting rokne ke liye **Dropout ($p = 0.5$)** invent kiya.

3. VGG-16 (2014, Oxford University)

- **3×3 Factorization Rule**:
    - Bade filters (5×5 ya 7×7) hata kar sirf chhote **3×3 filters ke stacks** use kiye.
    - **Kyun behtar hai?**: Do 3×3 filters milkar ek 5×5 filter jitna area dekhte hain, lekin parameters **28% kam** lagte hain aur do baar ReLU lagne se non-

linearity badh jati hai!

4. GoogLeNet / Inception (2014, Google)

- **Inception Module**: Multiple filter sizes ($1 \times 1, 3 \times 3, 5 \times 5$) ko alag-alag chunne ke bajaye parallel mein ek sath execute kiya aur unke outputs ko jod diya.
- Parameters ko kam rakhne ke liye 1×1 **Bottleneck Convolutions** use kiye.

5. ResNet (2015, Kaiming He - Microsoft Research)

- **Vanishing Gradients Ki Problem**: Jab network 50-100 layers lamba hota hai, toh backpropagation mein gradients multiply hote-hote zero ban jate hain (Vanishing Gradients).
- **ResNet Ka Solution: Skip Connection (Residual Highway)**:

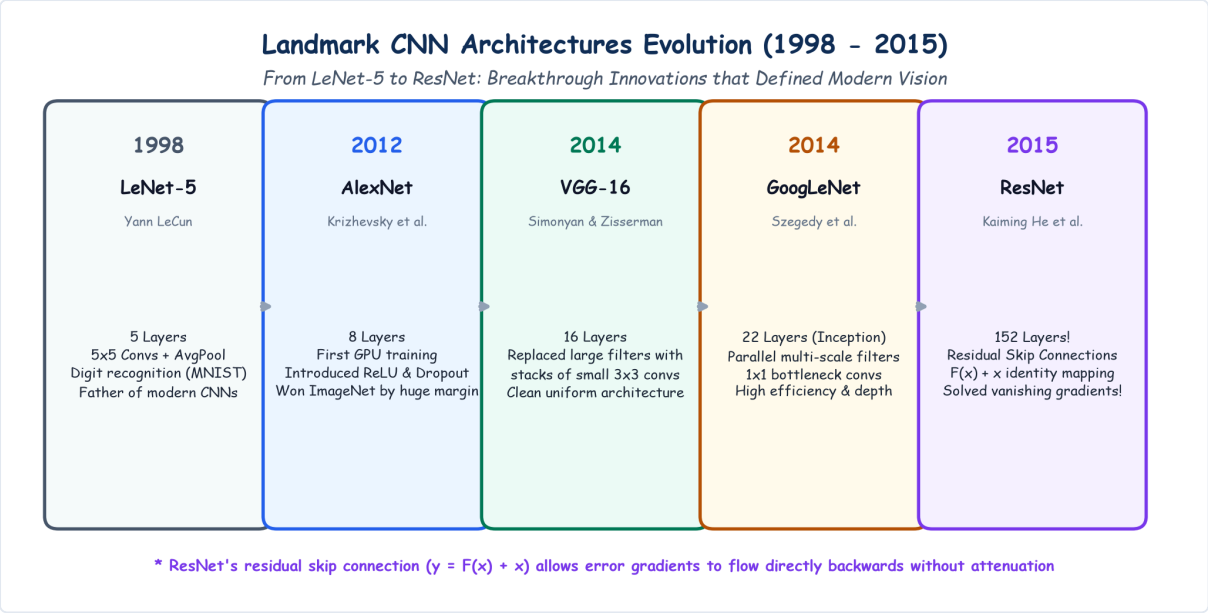
$$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}$$

  - Input $\mathbf{x}$ ko do layers ke upar se jump kara kar aage seedhe add kar diya jata hai.
  - Isse backpropagation ke dauran error gradients bina kisi attenuation ke seedhe pichli layers tak pahunchte hain.
  - Is invention ki wajah se **152 layers deep** network train ho paya aur isne pehli baar **Human Visual Accuracy (5%) ko maat de di (3.57% error)!**

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3. Hand-Drawn Educational Diagram: Architectural Evolution

Neeche diya gaya student-style handwritten diagram in 5 landmark models ke timeline aur unki key innovations ko visualize karta hai:



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4. Summary Table

Model	Saal	Layers	Sabse Badi Khoj	Error Rate
LeNet-5	1998	5	Pehla CNN digit reader	MNIST
AlexNet	2012	8	GPU + ReLU + Dropout	15.3%
VGG-16	2014	16	Small 3×3 filter stacks	7.3%
GoogLeNet	2014	22	Multi-scale Inception modules	6.7%
ResNet	2015	152	Residual Skip Connections ($F(x) + x$)	**3.57%** (Beats Humans!)